

Class 12 Extra Credit

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Population Scale Analysis

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

Q. How many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

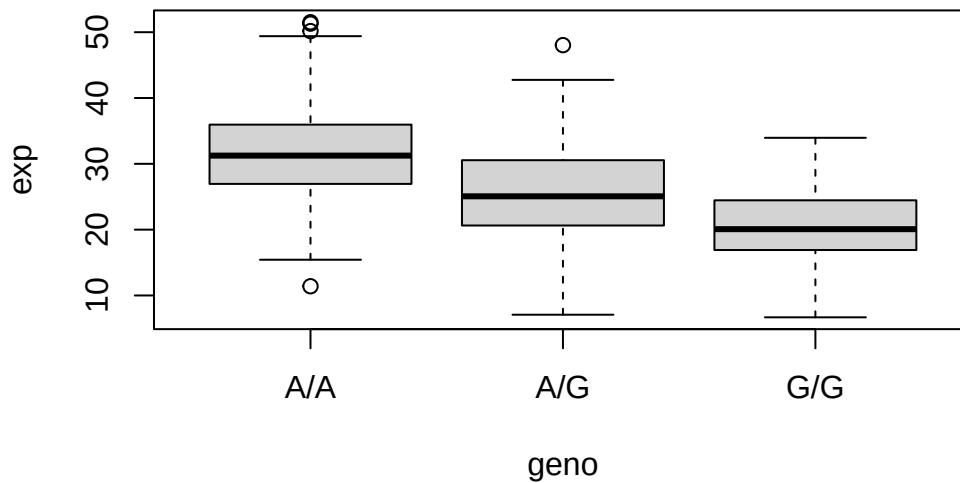
```
[1] 462
```

Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
box_plot <- boxplot(exp ~ geno, data=expr)
```



```
box_plot$stats[3,]
```

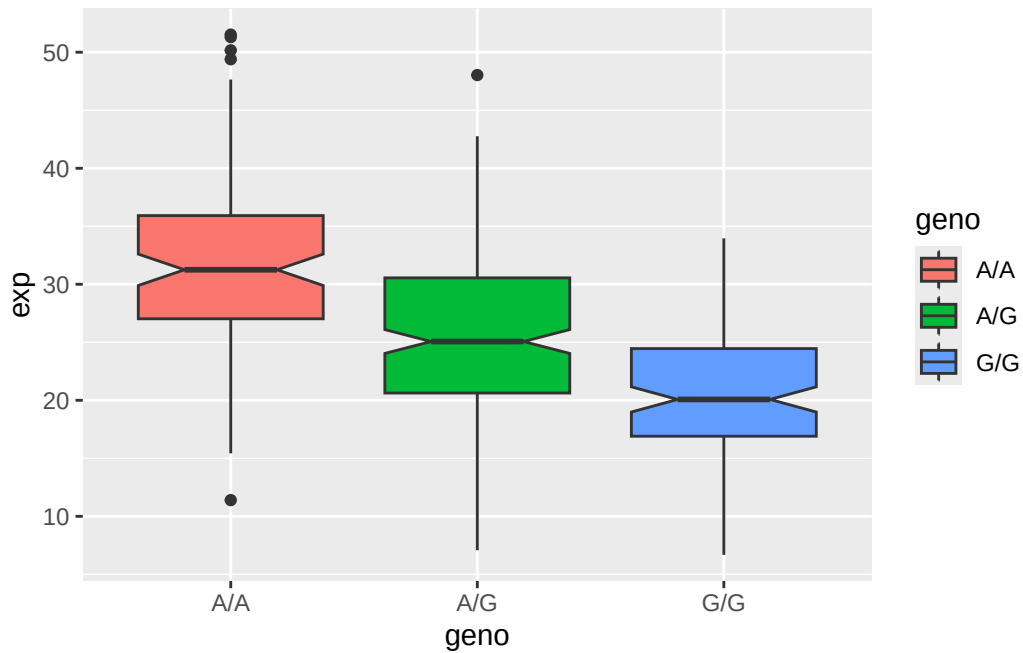
```
[1] 31.24847 25.06486 20.07363
```

- The sample size for A/A is 108, A/G is 233 and G/G is 121. The medians are 31.24847, 25.06486 and 20.07363 respectively.

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

```
ggplot(expr) + aes(x=geno, y=exp, fill=geno) + geom_boxplot(notch=T)
```



- From this plot, we can see that as the number of G alleles increases, the expression decreases so it can be inferred that ORM DL3 expression is highest in A/A, medium in A/G and lowest in G/G.
- This pattern wherein G allele increase is associated with decrease in expression suggests that the SNP does affect ORM DL3 expression in an allele-dependent way.